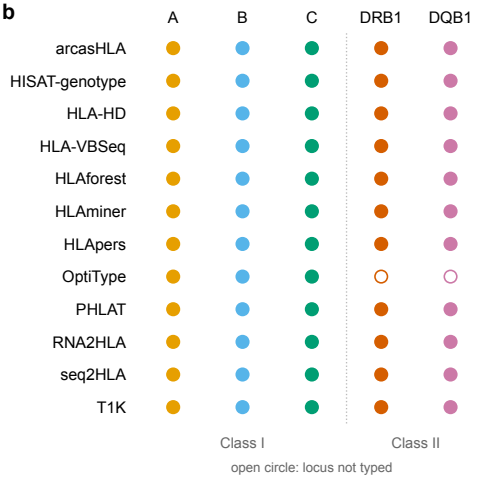
a

Dataset	n	Cells	Protocol	Reads (bp)	Gold standard
D1	490	LCL	bulk	2 x 75	Sanger SBT (1000 Genomes)
D2	86	PBMC	bulk	2 x 126	published typing
D3	50	LCL	bulk	2 x 37	published typing
D4	14	CD4+ T cells	bulk	2 x 126	published typing
D5	8	PBMC	bulk	2 x 38	published typing
D6	4	PBMC	bulk	2 x 37	published typing
D7*	20	Bone marrow	10x 3' scRNA	10x	NGS amplicon typing
D8*	3	Whole blood	bulk	in-house	PacBio HiFi assembly

* outside the D1-D6 main scope (partial caller coverage or n = 3)

b



c

Gold standard (D1 sample ERR188023, HLA-B)

`B*13:02  B*56:01/B*56:24`

an entry with alternatives ("/") matches any of them

	Call 1	Call 2	Outcome (2-field names compared)	Accuracy
T1K	`B*13:02:01 -> B*13:02`	`B*56:01:01 -> B*56:01`	2 correct	**2 / 2**
seq2HLA	`B*55:02`	`B*13:02`	1 correct, 1 miscalled	**1 / 2**
PHLAT	`B*13:03`	`B*55:50`	2 miscalled	**0 / 2**
HLAminer	`no call`	`no call`	2 no call	**0 / 2**

Accuracy = correct calls / all calls (two per sample and locus; no-calls count in the denominator). Calls are truncated to 2 fields (protein) or 1 field (allele group) before names are compared.